

Hospital Management & Appointment System

2520090016

P. Rahul

2520080005

A. Lakshmi Sai Teja

Project Abstract

- A single web-based platform that brings hospital operations together for patients and staff
- Today, registration, appointments, records, billing, pharmacy and lab work often run on separate or manual processes
- This causes delays, errors and poor coordination between departments
- Patients can register, book appointments, and track status online
- Doctors manage schedules and access Electronic Medical Records (EMR)
- Goal: reduce paperwork, save time, minimise errors, and improve coordination



One platform. Every department.

Built with React.js, Node.js/Express, MySQL (Sequelize), JWT authentication and Twilio notifications.

Designed for a simple, convenient experience for patients, doctors and administrators.

MAJOR AREAS COVERED



Patient Registration



Appointment Booking



Medical Records



Billing



Pharmacy



Laboratory Reports



Token / Queue Mgmt



Admin Management

Problem Statement



Manual paperwork



Long waiting times



Appointment management difficulties



Data entry errors



Difficulty accessing patient information



Poor coordination between departments



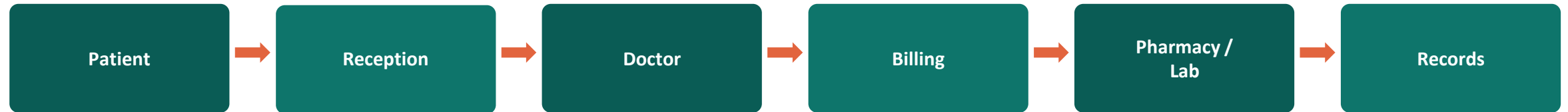
Separate handling of billing, pharmacy,
lab & records



Difficulty tracking queues and
appointments

CURRENT APPROACH

Existing System

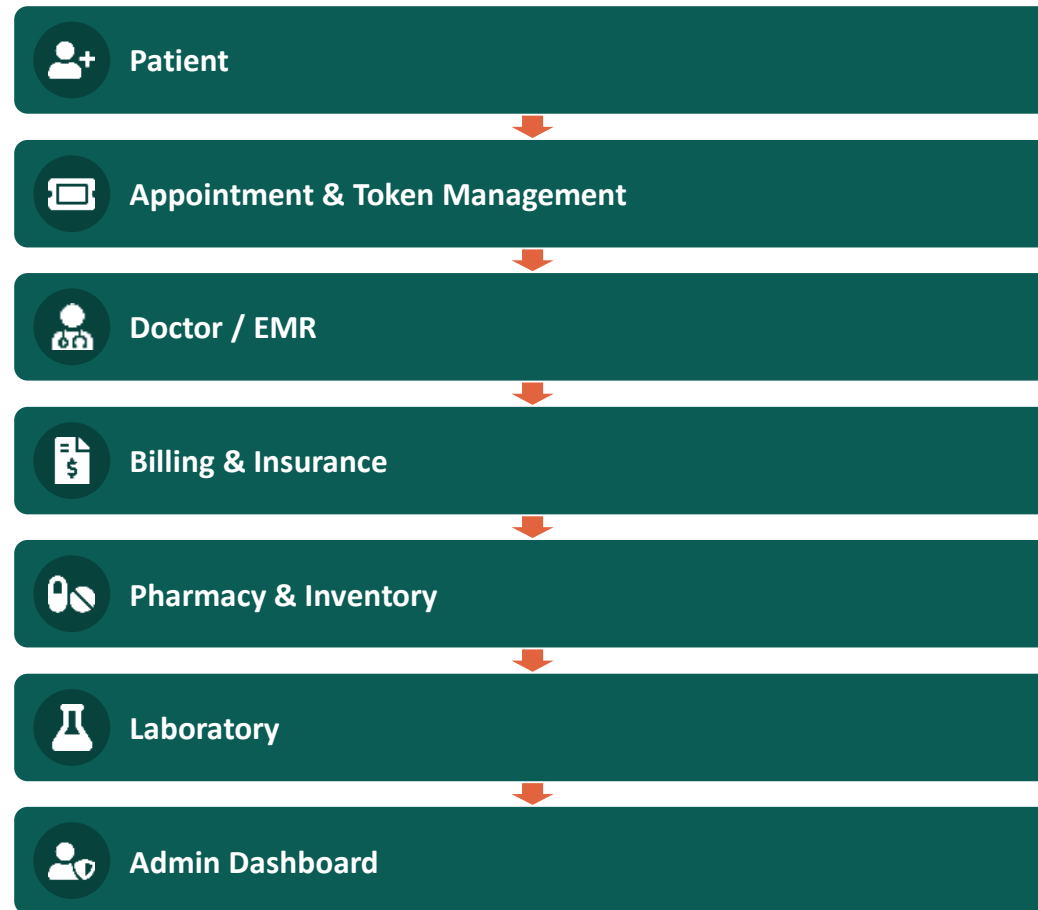


Each stage is handled separately, often with paper records passed manually between departments.

LIMITATIONS OF THIS APPROACH



Proposed System



A single, connected platform

- All major hospital operations run through one integrated system
- Patients register, book appointments and track status in real time
- Doctors access Electronic Medical Records (EMR) instantly
- Token & queue management reduces waiting time
- Billing, pharmacy and laboratory data stay connected to patient records
- Admin dashboard gives staff a centralised view of hospital activity

Objectives



Simplify patient registration & profiles



Reduce patient waiting time



Minimise manual data entry errors



Improve coordination between departments



Enable real-time EMR access for doctors



Streamline appointment booking & scheduling



Centralise hospital administration & monitoring



Integrate pharmacy & inventory tracking



Deliver a convenient experience for all users

Scope



Patient registration & profiles



Appointment booking & tracking



Doctor EMR access



Billing & insurance



Pharmacy & inventory



Laboratory report management



Token / queue management



Admin dashboard & monitoring

Covers a single hospital / clinic — patient, doctor and admin roles only.

Technology Identification



React.js



Node.js & Express.js



MySQL & Sequelize ORM



Database Design



JWT Authentication



Dev Tools



Twilio



RESTful APIs

A modern, secure full-stack toolchain connecting every module.

System Architecture



Thank You